The total space of the circle code is contractible

The Univalent Foundations Program

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Editorial note. The counted statement is the author lem environment at source lines 665-667 and its proof environment opens on line 668, the next line; the pairing is by adjacency and the retained proof range 668-699 is the whole environment.

Editorial note. This source has no label-number dump entry for this label (newlabel thm:iscontr-s1cover is absent from the pinned dump), so no counter value is imposed and the counted lemma prints with the wrapper numbering; the original number is not claimed.

Editorial note. The author defers one verification with the sentence "This is easy by induction on z, using the definition of q_z.", quoted verbatim; the main construction is carried out in full, which is the project criterion for accepting a complete main argument with a routine verification left open.

Editorial note. Two labels the proof cites lie outside the card and are declared in external_references as non-load-bearing navigation pointers: thm:flattening (book Lemma 6.12.2, defined at line 1833 of the hits chapter pinned file) and thm:sign-induction (book Theorem 6.10.12, defined at line 1399 of the same file and counted by its own card of this lane).

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: contractibility of the total space of the circle code (author label thm:iscontr-s1cover).

Allowed standard prerequisites (editorial).

- The circle code and its total space. The code family assigns a type to each point of the circle, and the total space consists of a point together with an element of its code.. It is the type shown to be contractible.
- The flattening lemma. For a span whose pushout is a higher inductive type and a family with an equivalence datum, the total space of the induced family is again a higher inductive type with explicit constructors.. It rewrites the total space as a higher inductive type with a point constructor and paths between consecutive integers.
- Induction on the integers. A family over the integers with a point at zero and steps in both directions has a section.. It builds the paths from the centre to every point of the resulting type.

Source reference. Original lemma 6.12.2: sources/hott-hits/pinned-original-source.tex, source lines 1833–1833. Book Lemma 6.12.2 (the flattening lemma), applied at the very start of the retained proof with an explicit choice of span, family and equivalence; the application is carried out in the retained text and the pointer is navigation only.

Source reference. Original theorem 6.10.12: sources/hott-hits/pinned-original-source.tex, source lines 1399–1399. Book Theorem 6.10.12, the two-sided induction principle for the integers, invoked for the construction of the paths to all generators. Navigation pointer only: it is counted by its own card of this lane and the retained proof carries out the induction.

Lemma 0.0.1. *The type* $\sum_{(x:\mathbb{S}^1)} \mathsf{code}(x)$ *is contractible.*

Proof. We apply the flattening lemma (original lemma 6.12.2) with the following values:

- $A :\equiv \mathbf{1}$ and $B :\equiv \mathbf{1}$, with f and g the obvious functions. Thus, the base higher inductive type W in the flattening lemma is equivalent to $\mathbb{S}^1$.
- $C : A \to \mathcal{U}$ is constant at $\mathbb{Z}$.
- $D : \prod_{(b:B)}(\mathbb{Z} \simeq \mathbb{Z})$ is constant at succ.

Then the type family $P : \mathbb{S}^1 \to \mathcal{U}$ defined in the flattening lemma is equivalent to $\mathsf{code} : \mathbb{S}^1 \to \mathcal{U}$. Thus, the flattening lemma tells us that $\sum_{(x:\mathbb{S}^1)} \mathsf{code}(x)$ is equivalent to a higher inductive type with the following generators, which we denote R:

- A function $\mathsf{c} : \mathbb{Z} \to R$.
- For each $z : \mathbb{Z}$, a path $\mathsf{p}_z : \mathsf{c}(z) = \mathsf{c}(\mathsf{succ}(z))$.

We might call this type the **homotopical reals**; it plays the same role as the topological space $\mathbb{R}$ in the classical proof.

Thus, it remains to show that R is contractible. As center of contraction we choose $\mathsf{c}(0)$; we must now show that $x = \mathsf{c}(0)$ for all $x : R$. We do this by induction on R. Firstly, when x is $\mathsf{c}(z)$, we must give a path $q_z : \mathsf{c}(0) = \mathsf{c}(z)$, which we can do by induction on $z : \mathbb{Z}$, using original theorem 6.10.12:

$$\begin{aligned} q_0 &\coloneqq \mathsf{refl}_{\mathsf{c}(0)} \\ q_{n+1} &\coloneqq q_n \cdot \mathsf{p}_n && \text{for } n \geq 0 \\ q_{n-1} &\coloneqq q_n \cdot {\mathsf{p}_{n-1}}^{-1} && \text{for } n \leq 0. \end{aligned}$$

Secondly, we must show that for any $z : \mathbb{Z}$, the path q_z is transported along p_z to q_{z+1}. By transport of paths, this means we want $q_z \cdot \mathsf{p}_z = q_{z+1}$. This is easy by induction on z, using the definition of q_z. This completes the proof that R is contractible, and thus so is $\sum_{(x:\mathbb{S}^1)} \mathsf{code}(x)$. □